

GHG Assurance Graph: Inspectable provenance and bounded comparison of organizational emissions

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Abstract

GHG Assurance Graph is an offline Python library for inspecting organizational greenhouse-gas evidence without confusing reported totals with reconstructed calculations. Typed records, RDF provenance, selected executable constraints, bounded version comparison and portable evidence packages support reproducible review. A synthetic benchmark resolves 45 result lineages and detects ten seeded defects; annotation-assisted comparison reconciles nine expected components. These are public, nonblind regression results, not autonomous causal accuracy. Three public-disclosure workflows preserve 24 assertions from UMC and TSMC, with 144 checks not assessable and all causal drivers UNKNOWN. The software exposes missing evidence rather than manufacturing assurance, completeness or standards conformity.

Keywords: greenhouse gases; provenance; SHACL; reproducibility; change attribution; evidence graphs

Software metadata

N r	Code metadata description	Metadata
C 1	Current code version	0.3.0a1, qualified alpha
C 2	Permanent link to code/repository used for this code version	https://github.com/simon5530/ghg-assurance-graph/tree/b0de4b8a9373eee24d631cad14265e37bffc75d
C 3	Legal code license	MIT; upstream dependencies retain their licenses
C 4	Code versioning system used	Git
C 5	Software code languages, tools and services used	Python; local CLI/library; no hosted service required
C 6	Compilation requirements, operating environments and dependencies	Python 3.12; tested 3.12.14 on macOS arm64 and hosted Ubuntu 24.04; uv.lock; Pydantic, Pint, RDFLib, pySHACL 0.31.0, RO-Crate 0.15.1; package verification requires POSIX facilities
C 7	If available, link to developer documentation/manual	https://github.com/simon5530/ghg-assurance-graph/tree/v0.3.0a1/docs

1. Motivation and significance

Organizational greenhouse-gas inventories combine activity evidence, emission factors, characterization conventions, boundaries and review decisions. A plausible total does not identify these dependencies. A year-to-year decrease may reflect changed activity, a corrected input, a different factor or a reporting-boundary change. Researchers need to inspect those alternatives before interpreting a numerical trend. The GHG Protocol Corporate Standard, Scope 2 Guidance and Scope 3 Standard define distinct accounting contexts [1–3]; a software check cannot substitute for establishing their applicability.

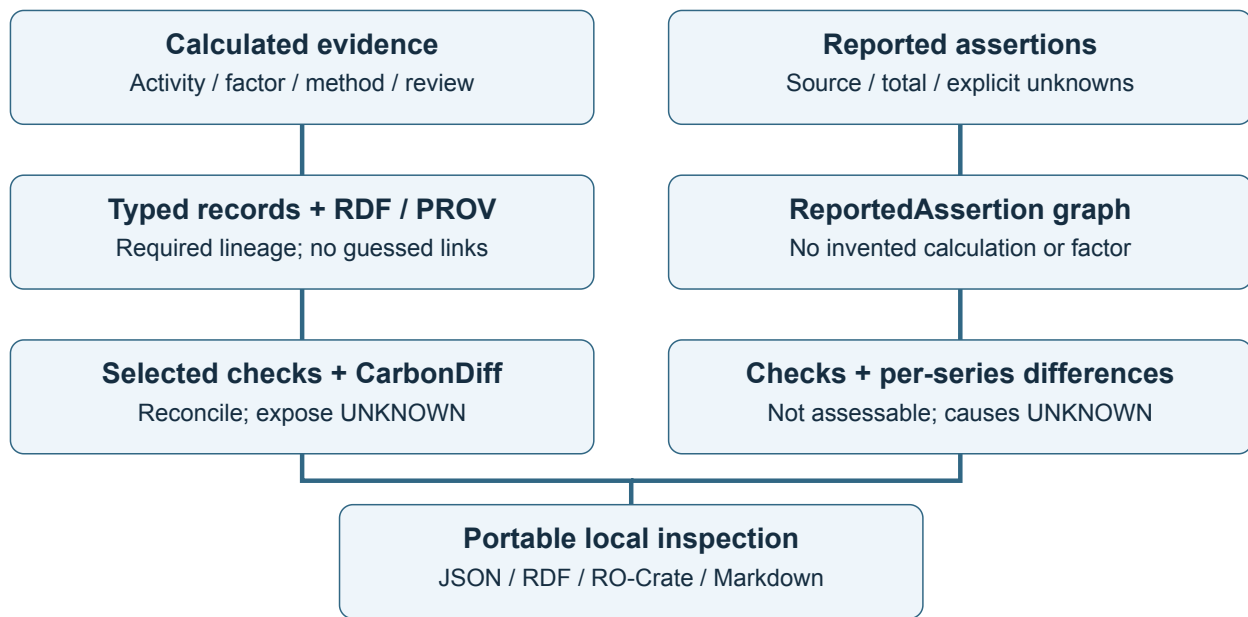
GHG Assurance Graph addresses a narrower problem: preserving inspectable evidence relationships, executing selected consistency checks and displaying what a comparison cannot explain. A researcher supplies local structured records, runs a command-line workflow, examines findings and exports a portable evidence directory. Two input profiles prevent an important category error. Calculated results require declared calculation lineage; public reported totals remain attributed assertions when that lineage is unavailable. Neither profile supplies an independent assurance opinion.

Existing infrastructure is reused rather than reimplemented. PROV-O represents provenance, SHACL supplies graph constraints, and RO-Crate describes portable research objects [4–6]. TEC Toolkit already combines carbon ontologies, PROV-based calculation provenance and executable validation [7]. CarbonLedger implements factor-vintage restatement [8]. Consequently, neither carbon provenance nor numerical factor-change attribution is claimed as new. The contribution is an experimental organizational-GHG integration with explicit authority boundaries, conservative missing-evidence behavior and reproducible examples. No comparative superiority has been measured; the repository's scoped GO rests on a same-case primary-source comparison of the remaining organizational integration contract, not universal novelty or practitioner endorsement.

2. Software description

2.1. Architecture and implementation

The implementation separates ingestion, canonical records, graph construction, validation, comparison and export (Figure 1). Pydantic rejects unknown fields and invalid references; Pint checks supported physical dimensions. RDFLib constructs collision-checked graphs and executes five allowlisted query templates. pySHACL runs packaged Core shapes with imports, advanced rules, JavaScript and inference disabled. Calculation and comparison do not require a language model, cloud credential, database or web service. Evidence URLs remain inert citations rather than instructions or automatically fetched inputs.



Outside software authority: authenticity / completeness / assurance

Figure 1. Implemented local workflows. Calculated evidence and reported assertions share inspection and export operations but not evidentiary status. Human source interpretation and assurance remain outside the computational boundary. Original schematic generated by the accompanying deterministic Python script; AI assisted its code and explanatory design. No source-report or standards artwork is reproduced.

The canonical schema has 17 record types covering organizations, facilities, periods, inventory versions, sources, activities, evidence, factors, GWP sets, methods, runs, results, boundaries, quality, reviews, changes and findings. Logical identifiers are distinct from revision identifiers. Conflicting content under one revision is rejected; no fuzzy identity matching silently joins inventories. Results link to generating calculations, calculations to used inputs, and reviews to reviewed targets. A retrospective review is not represented as generating the emissions result.

Calculated packages require explicit factor, method, GWP and boundary context. The bounded profile supports one activity and factor per run, not arbitrary multi-input life-cycle models. Generic JSON/CSV interchange accepts caller-mapped external results with complete provenance; it is not a tested PACT, openLCA or Brightway integration. Graph-derived Markdown provides an optional Obsidian inspection view without making Obsidian the database.

2.2. Validation and reported assertions

Validation combines closed schemas, reference integrity, selected SHACL constraints and independent raw-row domain rules. Findings have stable identifiers and explicit rule/target/severity fields. Checks include factor provenance, supported units, review metadata and arithmetic consistency. Raw decimal strings are checked using exact rational arithmetic, independently of the benchmark calculator. Some restrictions, such as the ACME factor-year policy, are deliberately narrower than accounting standards and must not be mistaken for universal requirements.

The additive reported-disclosure/1 profile stores organization, year, scope/category, Scope 2 basis, decimal-string quantity, source URL/hash/page/retrieval time and explicit boundary, GWP, restatement and rounding metadata. Unknown metadata remains null. A ReportedAssertion does not become an EmissionResult, ActivityRecord or CalculationRun. Validation distinguishes assessed,

invalid and not_assessable; the last is neither a defect nor a pass. Report-wide status remains not_assessable because totals alone cannot establish source authenticity, upstream recalculation or independent assurance.

Scope 2 location-based and market-based values are alternatives, never additive. Totals and their components are also not blindly summed. Public comparisons match explicit series and report arithmetic deltas separately from comparability. Missing years are missing, not zero. Where a total reconciliation is eligible, the tolerance uses disclosed rounding increments; an absent increment is not replaced with an invented tolerance.

2.3. CarbonDiff

CarbonDiff compares validated ACME raw snapshots through asserted namespace and stable row identifiers. For activity q , unallocated factor f and allocation share s , its ordered decomposition is:

- Activity component: $(q_1 - q_0) \times f_0 \times s_0$.
- Factor component: $q_1 \times (f_1 - f_0) \times s_0$.
- Allocation component: $q_1 \times f_1 \times (s_1 - s_0)$.

Interaction terms therefore belong to later operands. This is an explicit accounting convention, not uniquely identified causation. Bounded Decimal arithmetic isolates precision from the caller's context and enforces exact reconciliation: attributed change plus signed UNKNOWN residual equals the difference in totals. Absolute UNKNOWN exposure is reported separately so opposing unexplained changes cannot disappear through cancellation.

Unsupported substantive changes make the row's delta UNKNOWN unless a caller supplies a scoped semantic declaration covering the observed changes. Declarations contain evidence text and a label, not an expected amount. They are assertions requiring review, not authenticated causal evidence. Changed method, GWP or allocation policy must not silently become operational improvement. The synthetic calculator rejects unsupported market-based electricity, raw-gas characterization, offsets, removals and biogenic streams instead of manufacturing values; this refusal is a scope limit, not implementation of those accounting methods.

2.4. Portable evidence and reproducibility

The RO-Crate-based export contains canonical JSON, Turtle, JSON-LD, schema, vocabulary descriptor, metadata and a closed manifest. Explicit timestamps, software versions and commands support replay. Verification checks file membership, hashes, canonical payloads and regenerated exports without dereferencing remote contexts. Unexpected files, symlinks and inconsistent payloads are rejected. This verifies a bounded application profile, not every possible RO-Crate. A separately retained manifest digest can anchor integrity, but hashes alone do not authenticate an invoice, factor or assurance statement.

3. Illustrative examples

3.1. Synthetic ACME workflow

ACME Electronics is a fictional three-site organizational scenario, not a complete corporate inventory. Version 0.1 supplies three snapshots (2025-v1, 2025-v2, 2026-v1), each with 158 canonical records and 15 results. Factors and characterization bases are invented and nonproduction. Selected

Scope 3 sources are included; omitted applicable categories are not declared immaterial. The fixed seed is 20250926, and literal expected values plus a separate Fraction oracle check calculations.

From a locked installation, the principal commands are:

```
uv sync --locked
uv run ghgag benchmark run
uv run ghgag graph explain urn:ghgag:acme-2025-v1-electricity-result:v1 --input
benchmark/generated/2025-v1
uv run pytest -q
```

The electricity explanation resolves a 500 kgCO₂e result to its activity/evidence, factor, method, GWP, inventory, period, boundary and simulated review. Across the three clean snapshots, 45/45 required lineage chains resolve. This measures stored relationships, not whether underlying documents exist or tell the truth.

The partial totals are 3,820, 3,850 and 4,060 kgCO₂e. Ten separate defective fixtures produce TP=10, FP=0 and FN=0, hence precision, recall and F1 of 1.0. Development and publicly visible holdout subsets contain six and four defects respectively; three clean snapshots have no findings. Defects include duplicate activity, missing factor source, unsupported conversion and double GWP application. Because labels and scenarios are public and informed development, these scores are nonblind regression evidence, not estimates of unseen-inventory performance.

Annotation-assisted comparison reproduces nine expected component labels and amounts exactly, with zero mean absolute component error and zero residual. The restatement consists of a fleet correction (+20) and resolved travel data (+10). The next-year components are electricity activity (-100), electricity factor (+80), supplier mix (-100), method (-50), GWP (+400), boundary (+30) and allocation (-50), all kgCO₂e, summing to +210. Seven semantic labels are supplied assertions; only the two electricity components are mechanically decomposed. Without those declarations, the next-year signed UNKNOWN residual is +300 kgCO₂e and absolute UNKNOWN exposure is 500 (Figure 2). Zero assisted residual must not be described as autonomous causal accuracy.

Public synthetic regression evaluation — not blind generalization

Seeded findings: TP=10, FP=0, FN=0

Precision=1.00; recall=1.00; F1=1.00

Unassisted CarbonDiff (kg CO₂e):

2025-v1 → 2025-v2: delta 30.0; signed UNKNOWN 10.0; absolute UNKNOWN 10.0

2025-v2 → 2026-v1: delta 210.00; signed UNKNOWN 300.00; absolute UNKNOWN 500.00

UNKNOWN can cancel in the signed residual; absolute exposure is shown separately.

Figure 2. Executed ACME selected-defect regression and unassisted differences. The source generator reads committed fixtures and calls the evaluator; public labels are nonblind. Signed and absolute UNKNOWN quantities expose unresolved change rather than measured abatement. Original computational figure, with AI-assisted code; no generative image model or third-party artwork.

3.2. Public UMC and TSMC disclosures

A separate workflow tests whether real aggregate disclosures can be inspected without invented upstream evidence. A precommitted SHA256 seed selected UMC from the ordered candidates TSMC, UMC, Delta and ASE; TSMC provides a lightweight held-out transfer case, not a blinded or representative sample. Facts come from the official 2024-report vintages [9,10], rather than separate

annual-report vintages or the latest available disclosures. Source hashes and exact locators are retained. Extraction was checked against rendered tables; reproduction uses frozen facts, not a general PDF extraction model.

UMC Group contributes seven assertions, a separate parent-only fixture nine, and TSMC eight. Table 1 reports the numerical 2023–2024 differences. UMC parent and group boundaries are never merged. Group Scope 3 has only a selected 2024 value, 1,713,507 tCO₂e, so no prior-year change is invented. UMC's Scope 2 basis remains unspecified rather than inferred from narrative discussion.

Table 1. Selected reported 2023–2024 changes, tCO₂e. Source: UMC pp.99–100 and 209; TSMC p.266. Every cause is UNKNOWN; these are arithmetic differences, not comparable performance or operational attribution.

Series	UMC Group	UMC parent only	TSMC
Scope 1	–45,885	–69,518	+229,841
Scope 2, unspecified	–112,353	–42,158	Not supplied
Scope 2, location-based	Not supplied	Not supplied	+1,208,803
Scope 2, market-based	Not supplied	Not supplied	+770,010
Scope 3	No matched year	–476,241	+606,518

TSMC reports 2024 boundary additions, and its Scope 3 boundary descriptions conflict between pp.115 and 266. The conflict remains unresolved rather than silently corrected. Row-specific restatement and rounding information were not established for these series. Different boundaries, unavailable metadata and aggregate-only evidence prevent an unqualified like-for-like interpretation. Company-reported external verification is attributed disclosure, not an assurance engagement performed by this software.

All three runs are structurally accepted and their evidence packages verify internally, but all remain not_assessable overall. There are 33 assessed checks (three schema checks and 30 metadata-presence declarations), 144 not-assessable checks, zero invalid findings on the selected inputs, and upstream recalculation available for 0/24 assertions. Every causal driver remains UNKNOWN. The committed workflows contain 207 output files, including explanations, differences, validation, graphs, verified crates and inspection notes. Output-file counts describe artifact coverage, not additional observations or independent evidence.

3.3. Verification scope

The current regression baseline is 448 passing tests, separate from scientific performance metrics. Adversarial tests cover identity collisions, malformed inputs, precision isolation, unsupported methods, package tampering and denied writes. The synthetic detector does not receive the ground-truth file; the evaluator does. This separation reduces direct answer leakage but does not make development independent or blinded.

Public-case runners block Python socket connections and DNS during execution. Recorded same-machine isolated wheel installations reproduced UMC Group and TSMC output trees byte-for-byte after input timestamps were frozen. Installation still requires dependencies from a network or cache. A subsequent fresh GitHub-hosted Ubuntu runner independently fetched the published source, wheel and expected-output archive, verified pinned digests, installed separate environments and

reproduced all 207 output files byte-for-byte in both installations (run 36382755169). This is fresh-machine software reproduction, not a security sandbox or practitioner replication. Exact commands, environment and per-file digests are in the archived CI report.

4. Impact

The immediate utility is methodological transparency: a researcher can ask which factor revision supports a result, which evidence is missing, and whether a numerical change has sufficient context for interpretation. A second use is teaching the distinction between internally consistent data and a substantiated inventory. The public cases show that preserving “not assessable” can be more informative than constructing a superficially complete graph from missing inputs.

The software enables future studies of review effort, error discovery and disagreement over attribution conventions. It also supplies a reproducible starting point for comparing an evidence overlay with existing accounting and semantic-provenance tools. Those are prospective uses, not established outcomes. There are no measured independent users, adoption counts, commercial deployments, reductions in review time or publications using this software. Download statistics and repository activity are not substituted for research impact.

Important limitations remain. The model does not implement complete gas-resolved inventories, category-level uncertainty, organizational controls, full significance screening, production recalculation or independently authenticated review. ISO 14064-1:2018 source inspection informs a documented coverage assessment [11]; it does not establish conformity, and ISO categories must not be conflated with GHG Protocol scopes. Scope 2 contractual eligibility is not proved by retaining a market-based number. Synthetic policy restrictions do not become normative requirements. Broader matching, simultaneous semantic causes and vendor-specific adapters require further implementation and validation.

The authority boundary is deliberate: software may expose relationships and reconcile declared quantities, while qualified people must decide source truth, completeness, methodological appropriateness and assurance. Future evaluation should include independently specified cases, practitioner review and external reproduction before stronger claims are made.

5. Conclusions

GHG Assurance Graph provides an inspectable, offline evidence workflow connecting typed organizational-GHG records, semantic provenance, selected validation, bounded change comparison and portable outputs. Synthetic results demonstrate encoded contracts; public reports demonstrate useful preservation of missingness, not reconstructed emissions. The qualified alpha is a reproducible research artifact, not a certified accounting system or autonomous assurance provider. Human review and stronger impact evidence remain publication-development needs; source preservation and fresh-host software reproduction do not provide those forms of validation.

Code and data availability

MIT source and synthetic fixtures are available at <https://github.com/simon5530/ghg-assurance-graph>, with qualified prerelease vo.3.0a1 and frozen code commit in the metadata. Public numeric extracts and all 207 workflow outputs are under `examples/public_companies`; source citations identify the copyrighted company reports, which are not redistributed. Licensed standards text and

private evidence are excluded. Released source is preserved by Software Heritage snapshot swlh:1:snp:71149c7bca5b8439549ade1f2617371b5aa716bd; release object swlh:1:rel:cbd30c9cc797de3bcff750ece70c01167ea2bb23 resolves the tagged code. This preserves source-control objects, not all release binaries or a retention SLA. No archival DOI has been assigned. The manuscript's render script and original figure sources are included; the PDF is a reading copy, not the official submission format.

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Declaration of competing interest

Pending disclosure by every named author. A “no competing interests” declaration has not been authorized.

Declaration of generative AI and AI-assisted technologies

OpenClaw using openai/gpt-6-astra assisted software, tests, documentation, source inspection, manuscript drafting and deterministic figure-generation code. Earlier standards PDF inspection also used google/gemini-pro-latest, as recorded in the repository AI usage log. No model is required to reproduce the software results. Automated checks do not replace independent human verification. Human authors must review, revise and accept responsibility before submission; that approval has not yet occurred.

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